

CryptoPulse: Final Project Presentation

Alice Zheng, Bob Chen, Charlie Davis, Diana Lim

SUTD

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Recap: The Problem

- Predicting crypto prices is challenging due to high volatility.
- **Hypothesis:** Sentiment analysis provides a leading indicator for price movements.

- **Market Data (Quant):** Yahoo Finance (BTC-USD, ETH-USD).
- **Sentiment Data (Qual):**
 - ****News Sentiment:**** NLP analysis of headlines.
 - ****Fear & Greed Index:**** Real-time market emotion metric (0-100).

System Pipeline Architecture

- **1. Ingestion:** Yahoo Finance + News API + Fear & Greed.
- **2. Processing:** Merge $\rightarrow$ Fillna $\rightarrow$ Normalize (0 – 1).
- **3. Model (LSTM):**
 - Input: $[Price_t, Vol_t, Sent_t, FNG_t]$
 - Hidden: 2 Layers $\times$ 32 Units.
 - Output: $Price_{t+1}$
- **4. Application:** Streamlit Dashboard (Live Inference).

Enhanced User Interface (Live Demo)



Figure: CryptoPulse Market Intelligence Terminal (v2.0)

Experimental Results

Model	MSE	Improvement
Baseline (Naive)	0.051	-
Unimodal (Price)	0.042	+17.6%
Multimodal (Price + FNG)	0.032	+37.3%

Table: Multimodal Fusion significantly outperforms baseline models.

Conclusion & Future Work

1 Conclusion:

- Successfully fused quantitative (Price) and qualitative (Sentiment) data.
- Achieved **0.032 MSE**, a significant improvement over baseline.
- Delivered a premium, interactive "Market Intelligence Terminal".

2 Future Directions:

- Integrate **Transformer/Attention** mechanisms for better long-term forecasting.
- Connect to exchange APIs for **Automated Trading**.
- Incorporate real-time social media firehose (e.g., X/Twitter).

Thank You!

Project Repository:

<https://github.com/nizamkadirteach/cdsproject>